

Numerical Optimisation and Model Predictive Control A Hands-On Journey with CasADi and Impact CCTA 2026

M. Bos, L. Callens, W. Decré, A. Flórez, J. Gillis, E. Sandra, M. Schietecat, J. Swevers
August 11, 2026

Software overview

- All released as free and open-source software



CasADi

- self-contained symbolic framework for nonlinear programming
- algorithmic differentiation
- interfaces to own & third-party solvers
- C-code generation



Rockit

- optimal control prototyping
- free end-time & multi-stage problems



Impact

- from optimal control prototyping to NMPC deployment
- automatic generation of deployable artifacts



ImpaC++t

- closed-loop control in C++
- real-time scheduling & I/O
- simulation and deployment
- Linux RT kernel

- Additional software: Fatrop solver



Poll



< Activities



Visual settings



Edit



Join by Web Pollev.com/wilmdecre830

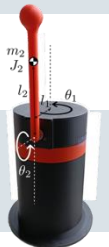


My experience with CasADi

Powered by  Poll Everywhere

Workshop overview

Session	Start	Topic
Le. 1	8:30 a.m.	Welcome & workshop overview Software installation Lecture CasADi – part 1
Br. 1	10:00 a.m.	Break
Le. 2	10:30 a.m.	Lecture CasADi – part 2 Exercise Alvik – static optimisation
Br. 2	12:00 p.m.	Lunch break
Le. 3	1:30 p.m.	Lecture MPC with Impact Exercise Alvik – dynamic optimisation
Br. 3	3:00 p.m.	Break
Le. 4	3:30 p.m.	Lecture ImpaC++t Exercise Furuta pendulum



Software installation

- Available at
<https://github.com/meco-group/CCTA-impact-2026#software-installation>



Questions?

